Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Midsem Exam, Date: 7 March, 2025

Timing: 15:00 to 16:30 | Spring 2025 | Max mark: 100

Solve all questions, each carry [20] marks.

Notation: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z) \equiv (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv (\partial_x, \partial_y, \partial_z) \equiv (\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi)$.
$A = |\vec{A}| = \sqrt{\vec{A}.\vec{A}}$ $\quad x \equiv x(t)$, $\dot{x} = \frac{d}{dt}x(t) \equiv Dx(t)$, $\ddot{x} = \frac{d^2}{dt^2}x(t) \equiv D^2x(t)$, $i = \sqrt{-1}$

1. **(a)** Express ∇ in (r, θ, ϕ)-coordinate system, where $x = r\cos\theta\sec\phi, y = r\sin\theta\sec\phi, z = r\tan\phi$. **[10]**
(b) Use Levi Civita symbols to prove $\vec{A} \times (\nabla \times \vec{A}) = \frac{1}{2}\nabla A^2 - (\vec{A}.\nabla)\vec{A}$ **[10]**

Ans. **(a)** To express the gradient operator ∇ in the given coordinate system, let's proceed systematically.

We have the transformation: $x = r\cos\theta\sec\phi, \quad y = r\sin\theta\sec\phi, \quad z = r\tan\phi$.. Our goal is to express the gradient operator ∇ in terms of (r, θ, ϕ)= $\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ in terms of (r, θ, ϕ). We first compute the derivatives of x, y, and z with respect to r, θ, and ϕ.

$$\frac{\partial x}{\partial r} = \cos\theta\sec\phi, \quad \frac{\partial y}{\partial r} = \sin\theta\sec\phi, \quad \frac{\partial z}{\partial r} = \tan\phi.$$

$$\frac{\partial x}{\partial \theta} = -r\sin\theta\sec\phi, \quad \frac{\partial y}{\partial \theta} = r\cos\theta\sec\phi, \quad \frac{\partial z}{\partial \theta} = 0.$$

$$\frac{\partial x}{\partial \phi} = r\cos\theta\sec\phi\tan\phi, \quad \frac{\partial y}{\partial \phi} = r\sin\theta\sec\phi\tan\phi, \quad \frac{\partial z}{\partial \phi} = r\sec^2\phi.$$

The scale factors h_r, h_θ, and h_ϕ are given by:

$$h_r = \sqrt{\left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2} = 1$$

$$h_\theta = \sqrt{\left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2} = r$$

$$h_\phi = \sqrt{\left(\frac{\partial x}{\partial \phi}\right)^2 + \left(\frac{\partial y}{\partial \phi}\right)^2 + \left(\frac{\partial z}{\partial \phi}\right)^2} = \frac{r}{\cos\phi}.$$

Using the formula for the gradient in an orthogonal curvilinear coordinate system we get,

$$\nabla = \hat{e}_r\frac{1}{h_r}\frac{\partial}{\partial r} + \hat{e}_\theta\frac{1}{h_\theta}\frac{\partial}{\partial \theta} + \hat{e}_\phi\frac{1}{h_\phi}\frac{\partial}{\partial \phi} = \hat{e}_r\frac{\partial}{\partial r} + \hat{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{e}_\phi\frac{\cos^2\phi}{r}\frac{\partial}{\partial \phi}.$$

(b) Using the Levi-Civita symbols: $(\nabla \times \vec{A})_i = \epsilon_{ijk}\partial_j A_k$.
Similarly, the left-hand side of the identity:

$$(\vec{A} \times (\nabla \times \vec{A}))_i = \epsilon_{ilm}A_l(\nabla \times \vec{A})_m = \epsilon_{ilm}A_l\epsilon_{mjk}\partial_j A_k = (\delta_{ij}\delta_{lk} - \delta_{ik}\delta_{lj})A_l\partial_j A_k = \delta_{ij}A_l\partial_j A_l - \delta_{ik}A_l\partial_l A_k.$$

where in the last step we use the Levi-Civita identity:$\epsilon_{ilm}\epsilon_{mjk} = \delta_{ij}\delta_{lk} - \delta_{ik}\delta_{lj}$.
Substituting the expression for $(\nabla \times \vec{A})_m$:

$$(\vec{A} \times (\nabla \times \vec{A}))_i = \epsilon_{ilm}A_l\epsilon_{mjk}\partial_j A_k.$$

Since $\delta_{ij} A_l \partial_j A_l = A_l \partial_i A_l$, and relabeling indices: $(\vec{A} \times (\nabla \times \vec{A}))_i = A_j \partial_i A_j - A_l \partial_l A_i$.

Finally, recognizing the terms:

$$A_j \partial_i A_j = \frac{1}{2}\partial_i (A_j A_j) + \frac{1}{2}(A_j \partial_i A_j - A_j \partial_j A_i),$$

where the second term is antisymmetric and vanishes under contraction, we obtain:

$$\vec{A} \times (\nabla \times \vec{A}) = \frac{1}{2}\nabla(A^2) - (\vec{A} \cdot \nabla)\vec{A}.$$

2. **(a)** Find an analytic function $f(z)$ whose real part is $x^2 + xy - y^2$. **[10]**
(b) Find regions in complex plane where both $f_1(z) = |y| - ix$ and $f_2(z) = |x| + i|y|$ are analytic. **[10]**

Ans. **(a)** $u = x^2 + xy - y^2$, $\frac{\partial(u)}{\partial(x)} = 2x + y = \frac{\partial(v)}{\partial(y)}$ then $v(x, y) = 2xy + \frac{y2}{2} + C_1$, similarly $\frac{\partial(u)}{\partial(y)} = x - 2y = -\frac{\partial(v)}{\partial(x)}$ then $v(x, y) = -\frac{x^2}{2} + 2xy + c_2$, so final $v(x, y) = -\frac{x^2}{2} + \frac{y^2}{2} + 2xy + c$

(b) $f_1(z) = |y| - ix$ from C-R equation $\frac{\partial(u)}{\partial(y)} = -\frac{\partial(v)}{\partial(x)}$ so it is only possible when $x > 0$ (1st and 4th quadrant)
from $f_2(z) = |x| + i|y|$ from C-R equation $\frac{\partial(u)}{\partial(x)} = \frac{\partial(v)}{\partial(y)}$ so it is only possible when $x > 0$ and $y > 0$ (1st quadrant).

Thus both functions are analytic only in first quadrant.

3. Find the unit vectors which are parallel to the tangent line of the curves $\Re(ze^z)$ and $\Im(ze^z)$ at $z = z_0 = 1 + i$, and show that they are orthogonal. Use Cauchy–Riemann criteria to show that $z_0 = 1 + i$ is not special, the orthogonality holds at any other arbitrary point $z_0 = a + ib$. **[20]**

Ans.

$$\begin{aligned} ze^z &= (x + \ iy)\ e^{x+iy} \\ &= (x + \ iy)\ e^x\ e^{iy} \\ &= (x + \ iy)\ e^x\ (\cos(y) + \ i\sin(y)) \\ &= e^x\ ((x\cos(y) - \ y\sin(y)) + \ i(x\sin(y) + \ y\cos(y))) \end{aligned}$$

Hence, we get:

$$\Re(ze^z) = e^x\ (x\cos(y) - \ y\sin(y)) = u(x, y) \quad \textit{and} \quad \Im(ze^z) = e^x\ (x\sin(y) + \ y\cos(y)) = v(x, y).$$

In order to find the unit vectors parallel to the tangent lines of these two curves at any point in $\mathbb{R}^3$, we must compute the gradients of the respective functions first. We get the following:

- $$\begin{aligned} \vec{\nabla}(\Re(ze^z)) &= \vec{\nabla}(e^x\ (x\cos(y) - \ y\sin(y))) \\ &= (e^x\ ((x+1)\cos(y) - \ y\sin(y)),\ -e^x\ ((x+1)\sin(y) + \ y\cos(y)),\ 0) \end{aligned}$$

- Similarly:

$$\begin{aligned} \vec{\nabla}(\Im(ze^z)) &= \vec{\nabla}(e^x\ (x\sin(y) + \ y\cos(y))) \\ &= (e^x\ ((x+1)\sin(y) + \ y\cos(y)),\ e^x\ ((x+1)\cos(y) - \ y\sin(y)),\ 0) \end{aligned}$$

For, $z = z_0 = 1 + i$, we have:

$$\vec{\nabla}(\Re(ze^z))\Big|_{z=z_0} = (e\ (2\cos(1) - \ \sin(1)),\ -e\ (2\sin(1) + \ \cos(1)),\ 0),$$

$$\textit{and} \qquad \vec{\nabla}(\Im(ze^z))\Big|_{z=z_0} = (e\ (2\sin(1) + \ \cos(1)),\ e\ (2\cos(1) - \ \sin(1)),\ 0).$$

Since the unit vectors parallel to the respective curves are normal to these above obtained vectors, and these vectors lie in $\mathbb{R}^2$, it is sufficient to show that these vectors are normal to each other. A straightforward computation shows that the inner product between these vectors is indeed 0. The explicit unit vectors are as follows:

- $\Re(ze^z)\Big|_{z=z_0}$:
$$\pm\frac{1}{\sqrt{5}}\ (2\sin(1) + \ \cos(1),\ 2\cos(1) - \ \sin(1),\ 0).$$
- $\Im(ze^z)\Big|_{z=z_0}$:
$$\pm\frac{1}{\sqrt{5}}\ (2\cos(1) - \ \sin(1),\ -(2\sin(1) + \ \cos(1)),\ 0).$$

It is evident from the expressions of gradients of $\Re(ze^z)$ and $\Im(ze^z)$ that Cauchy-Riemann condition is satisfied by the given function ze^z, and the orthogonality holds, for all $x, y \in \mathbb{R}$.

Another way. $z_0 = 1 + i \equiv (x_0, y_0) = (1, 1)$ We need contour lines of $u(x, y)$ and $v(x, y)$ that pass through $(x_0, y_0) = (1, 1)$. They are defined by the equations,

$$u(x, y) = u(1, 1) \ \ \textit{and} \ \ v(x, y) = u(1, 1)$$

Differentiating wrt x : $\frac{du}{dx} = 0, \frac{dv}{dx} = 0$, which implies (with $y' \equiv \frac{dy}{dx}$

$$\frac{du}{dx} = 0 \implies -e^x\{y(\sin y + y'\cos y) + (1 + x)(-\cos y + y'\sin y)\} = 0$$

$$y' = \frac{(1 + x)\cos y - y\sin y}{(1 + x)\sin y + y\cos y} \equiv m_u$$

y' is the slope m_u of the tangent at any point (x, y) on the curve $u(x, y) = u(1, 1)$. Similarly,

$$\frac{dv}{dx} = 0 \implies -e^x\{((1 + x)(\sin y + y'\cos y) + y(-\cos y - y'\sin y)\} = 0.$$

And solving for y' we get the slope of the tangent $m_v$to $v(x, y) = v(1, 1)$ at any point (x, y) as

$$m_v = -\frac{(1 + x)\sin y + y\cos y}{(1 + x)\cos y - y\sin y}$$

Since $m_u m_v = -1$, the the tangent lines are perpendicular to each other at any point (x, y).

4. **(a)** Use variable transformation to convert the equations $\{\dot{x} = x\ln y; \dot{y} = -y\ln x\}$ to linear form and solve them using matrix method for initial condition $x(0) = e^2$ and $y(0) = 1/e$ **[10]**

   (b) Show that particular integral of the diff eqn $(D - a)x(t) = f(t)$ is $e^{at}\int\limits_0^t f(t')e^{-at'}dt'$. **[10]**

Ans. **(a)** $u = \ln x, v = \ln y \implies \dot{u} = \frac{\dot{x}}{x}, \dot{v} = \frac{\dot{y}}{y}$. Thus, the given differential Eqs can be written as $\dot{u} = v, \dot{v} = -u$ with initial conditions $u(0) = 2, v(0) = -1$. The solution is then $\begin{bmatrix} u \\ v \end{bmatrix}_t = e^{tM} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, where $M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Since $Tr(M) = 0, \det(M) = 1$, $\lambda_\pm = \pm i$,

with eigenvectors $|e_\pm\rangle = \begin{bmatrix} 1 \\ \lambda_\pm \end{bmatrix} = \begin{bmatrix} 1 \\ \pm i \end{bmatrix}$.*(How to calculate? see Q5(a))*,

Initial vector in the eigenbasis of M : $\begin{bmatrix} 2 \\ -1 \end{bmatrix} = c_+|e_\pm\rangle + c_+|e_-\rangle = \begin{bmatrix} c_+ + c_- \\ i(c_+ - c_-) \end{bmatrix} \implies c_\pm = 1 \pm \frac{i}{2}$.

Finally $\begin{bmatrix} u \\ v \end{bmatrix}_t = e^{tM} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = c_+ e^{it}|e_+\rangle + c_- e^{-it}|e_-\rangle = \begin{bmatrix} 2\cos t - \sin t \\ -2\sin t - cost \end{bmatrix}$

Thus $x(t) = e^{u(t)} = e^{2\cos t - \sin t}, y(t) = e^{v(t)} = e^{-2\sin t - cost}$.

(b) Say, $y(t) = x(t)e^{-at}$. Then $\dot{y} = \dot{x}e^{-at} - axe^{-at} = e^{-at}(D - a)x = e^{-at}f(t)$. Thus, $y(t) = \int_0^t dt' e^{-at'} f(t')$. Since $y(t) = x(t)e^{-at}$, we get $x(t) = e^{at} \int_0^t f(t')e^{-at'} dt'$.

5. **(a)** Given a matrix $M = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, compute M^n **[12]** **(b)** What is $\lim_{n\to\infty} M^n$? **[8]**

Ans: **(a)** $Tr(M) = 6/5, \det(M) = \frac{1}{5}$; $\lambda_\pm = \frac{1}{2}[Tr \pm \sqrt{Tr^2 - 4Det}] = \frac{1}{2}(\frac{6}{5} \pm \frac{4}{5}) = 1, \frac{1}{5}$. $\implies D = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$

Eqn for eigenvectors $|e_\pm\rangle = \begin{bmatrix} 1 \\ a_\pm \end{bmatrix}$: $\frac{1}{5}\begin{bmatrix} 2 - 5\lambda_\pm & 1 \\ 3 & 4 - 5\lambda_\pm \end{bmatrix}\begin{bmatrix} 1 \\ a_\pm \end{bmatrix} = 0 \implies a_\pm = 5\lambda_\pm - 2 = 3, -1$.

Thus $|e_+\rangle = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $|e_-\rangle = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies U = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}, \det(U) = -4 \implies U^{-1} = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}$

Since $MU = UD$, $M = UDU^{-1}$, and

$$M^n = UD^nU^{-1} = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} 1^n & 0 \\ 0 & (\frac{1}{5})^n \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 + \frac{3}{5^n} & 1 - \frac{1}{5^n} \\ 3 - \frac{3}{5^n} & 3 + \frac{1}{5^n} \end{bmatrix}$$

(b) $\lim_{n\to\infty} M^n = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$ all $\frac{1}{5^n}$ terms vanish.